INCH-POUND

8 April 1993
SUPERSEDING
MIL-C-55302/65F
26 August 1988

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES
RECEPTACLE, SOCKET CONTACTS, DECADE INCREMENTS 10 THRU 70 CONTACT POSITIONS AND
INTERMEDIATE POSITIONS OF 14, 24, 44, 54, AND 26, 36, 56, AND 66 FOR
PRINTED WIRING BOARDS (.100 INCH SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55302

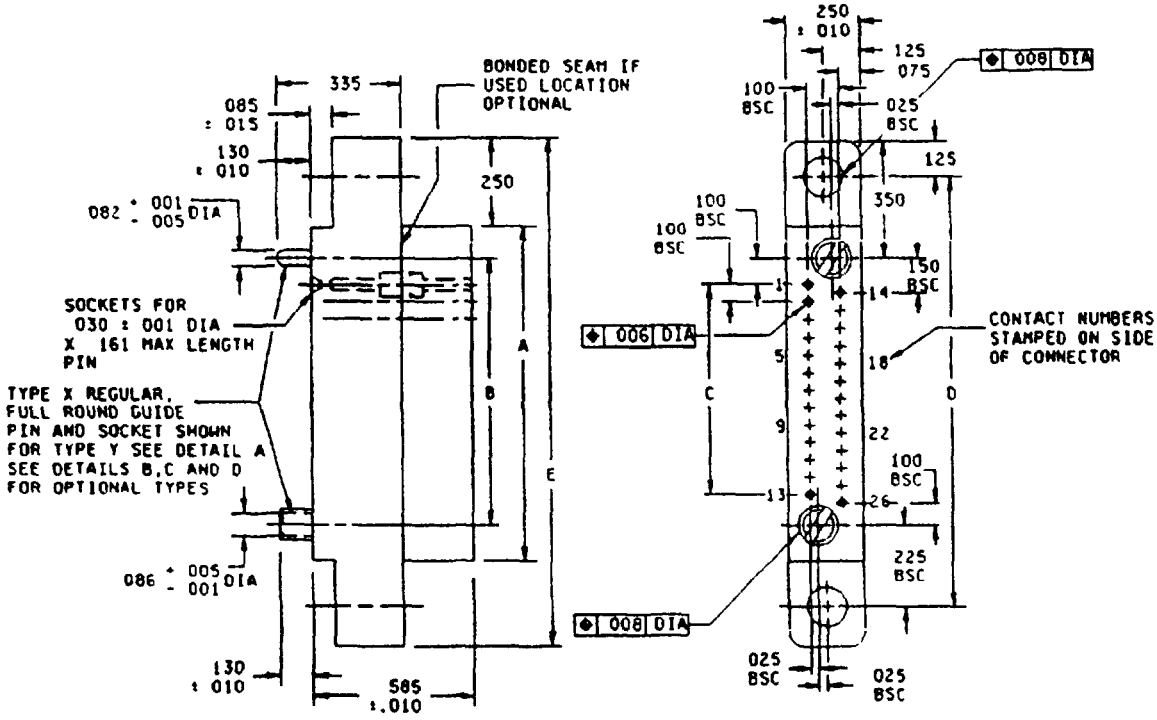
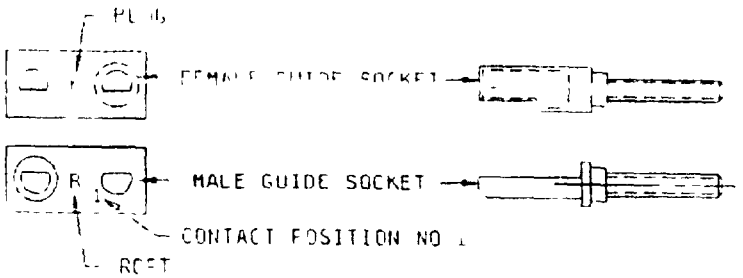


FIGURE 1. Connector, receptacle, .100 (2.54 mm) spacing

(C) Denotes changes

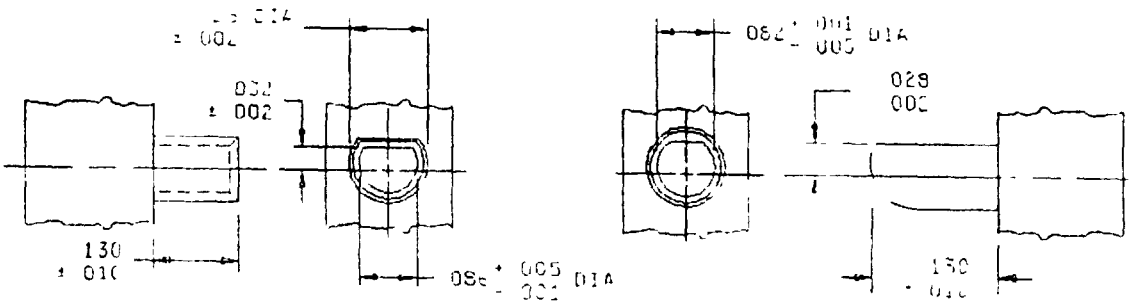
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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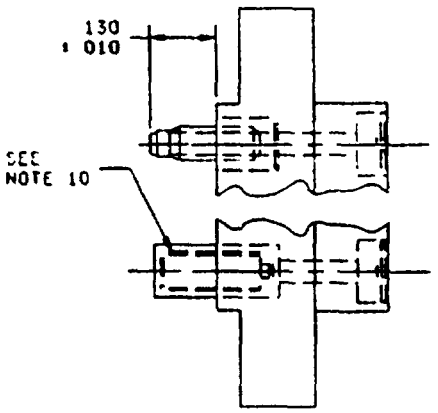
CONNECTOR RECEPTACLE WITH [illegible]

DETAIL A



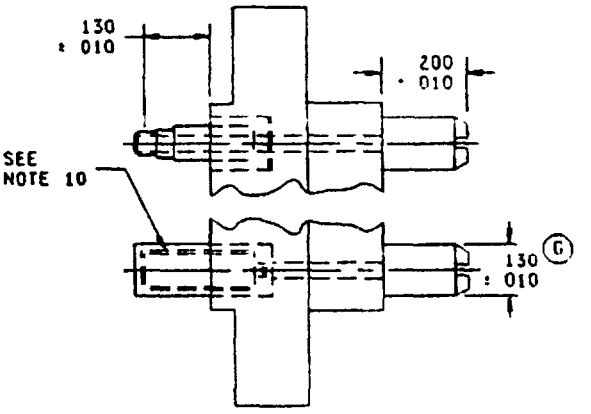
[illegible] Connector, receptacle, .100 (2.54 mm) [illegible] (for board

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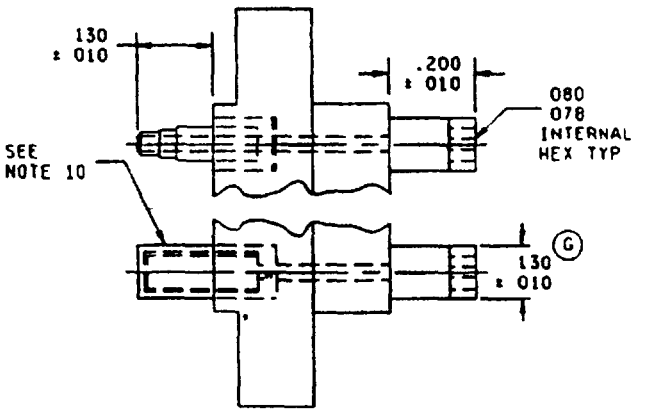
TYPE F FIXED JACKSCREWS

DETAIL B



TYPE S SHORT TURNING SLOTTED JACKSCREWS

DETAIL C



TYPE H SHORT TURNING HEX JACKSCREWS

DETAIL D

FIGURE 1 Connector, receptacle, 100 (2.54 mm) spacing - Continued

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FIGURE 1 Connector, receptacle 100 (2 54 mm) spacing - Continued

Inches	mm
001	0 03
002	0 05
005	0 13
006	0 15
008	0 20
010	0 25
025	0 64
028	0 71
030	0 76
032	0 81
075	1 91
078	1 98
082	2 08
086	2 18
094	2 39
100	2 54
120	3 05
125	3 18
130	3 30
150	3 81
161	4 09
200	5 08
225	5 72
250	6 35
335	8 51
350	8 89
585	14 86

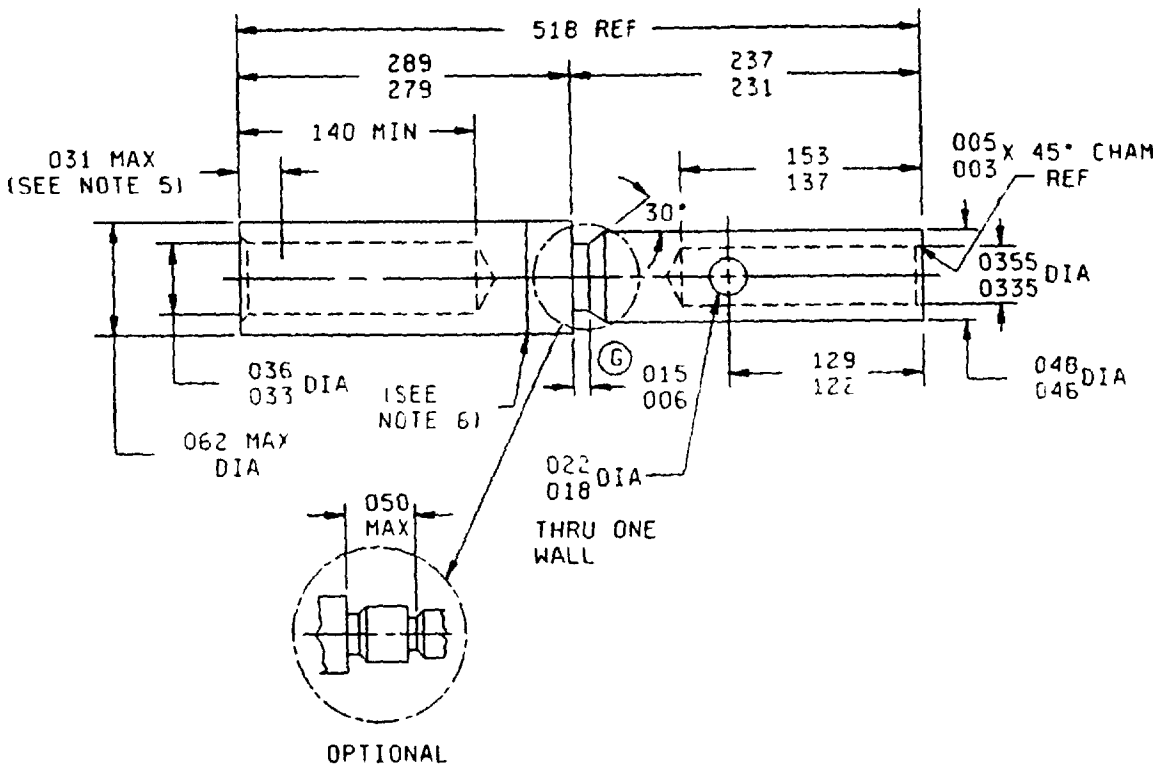
Number of contacts	Dimensions				
	A ± 010	BSC B	REF C	BSC D	E ± 010
10	850 (21 59)	650 (16 51)	400 (10 16)	1 100 (27 94)	1 350 (34 29)
14	1 050 (26 67)	850 (21 59)	600 (15 24)	1 300 (33 02)	1 550 (39 37)
20	1 350 (34 29)	1 150 (29 21)	900 (22 86)	1 600 (40 64)	1 850 (46 99)
24	1 550 (39 37)	1 350 (34 29)	1 100 (27 94)	1 800 (45 72)	2 050 (52 07)
26	1 650 (41 91)	1 450 (36 83)	1 200 (30 48)	1 900 (48 26)	2 150 (54 61)
30	1 850 (46 99)	1 650 (41 91)	1 400 (35 56)	2 100 (53 34)	2 350 (59 69)
36	2 150 (54.61)	1 950 (49 53)	1 700 (43 18)	2 400 (60 96)	2 650 (67 31)
40	2 350 (56 69)	2 150 (54 61)	1 900 (48 26)	2 600 (66 04)	2 850 (72 39)
44	2 550 (64 77)	2 350 (59 69)	2 100 (53 34)	2 800 (71 12)	3 050 (77 47)
50	2 850 (72 39)	2 650 (67 31)	2 400 (60 96)	3 100 (78 74)	3 350 (85 09)
54	3 050 (77 47)	2 850 (72 39)	2 600 (66 04)	3 300 (83 82)	3 550 (90 17)
56	3 150 (80 01)	2.950 (74.93)	2 700 (68 58)	3.400 (86 36)	3 650 (92 71
60	3 350 (85 09)	3.150 (80 01)	2 900 (73 66)	3 600 (91 44)	3 850 (97 79)
66	3 650 (92 71)	3 450 (87 63)	3 200 (81 28)	3 900 (99 06)	4 150 (105 41)
70	3 850 (97 79)	3 650 (92 71)	3 400 (86 36)	4 100 (104 14)	4 350 (110 49)

NOTES

1 Dimensions are in inches
2 Metric equivalents are given for general information only
3 Unless otherwise specified, tolerance is ± 005 (0 13 mm)
4 These connectors mate with connectors specified in MIL-C-55302/57, /61, and /63 using appropriate hardware
5 Numbers indicating the first and last position in each row and every fourth contact position in between shall be marked on the side(s) of the connector As an option to the above, numbers indicating every fourth cavity may be stamped on the side of the connector, with the exception that the number one contact shall be marked
6 Termination layout on 025 (0 64 mm) modular grid
7 Metric equivalents are in parentheses
8 Conformal coating clearances shall be 005 (0 13 mm) minimum (not required for optional design)
9 Optional design conformal coating clearance
(G) 10 Hole depth to female guide hardware 282 min (7 16 mm) Full thread depth to female threaded hardware 240 min (6 1 mm)

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FIGURE 1 Connector, receptacle .100 (2.54 mm) spacing - Continued



Inches	mm	Inches	mm	Inches	mm
.002	0 05	031	0 79	128	3 28
.003	0 08	033	0 84	137	3 48
004	0 10	0335	0 851	153	3 89
005	0 13	0355	0.902	166	4 22
.006	0 15	036	0 91	231	5 87
.011	0 28	046	1 17	237	6 02
018	0 46	048	1 22	279	7 09
022	0 56	062	1 57	289	7 34
.029	0 74	.122	3.10	518	13 16

NOTES
1. Dimensions are in inches
2. Machined surfaces to be 63 microinches or less
3. Remove all burrs and sharp edges .003 maximum
4. All diameters to be concentric within .004 T I R
5. Depth at which a square ended pin of .029 diameter first engages the socket contact spring
6. Maximum gap of .002 between hood and body
7. Contact part numbers shall be M55302/65-01 for standard force and M55302/65-02 for low insertion force contacts

FIGURE 2 Socket, contact, crimp

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REQUIREMENTS

Dimensions and configuration See figure 1

(G) Material Guide pins, type X guide sockets, jackscrews and jacksockets shall be made of corrosion-resisting stainless steel in accordance with ASTM-A581 or ASTM-A582 passivated in accordance with QQ-P-35 or shall be corrosion-resistant copper nickel alloy (61 ±2% copper, 25 ±2% nickel, 2 5% max other, and the balance zinc) and shall have a brinell hardness of 145 to 175 Type Y guide sockets shall be beryllium copper in accordance with ASTM B196 or ASTM B197, nickel plated in accordance with QQ-N-290, class 2, 50 to 150 microinches

Plating The contact plating shall be in accordance with MIL-C-55302 or gold in accordance with MIL-G-45204, type II, grade C, class 1, over nickel plating in accordance with QQ-N-290, class 2, 50 to 150 microinches in the engaging area and gold in accordance with MIL-G-45204, type II, grade C, class 00, or type III, grade A, class 00, over nickel plating in accordance with QQ-N-290, class 02, 50 to 150 microinches in the termination area

Contact identification See figure 2

Contact engagement and separation forces (No 22 contacts)

Minimum separation 0 5 ounce

Maximum engagement 12 0 ounces for standard force and 4 0 ounces for low force

Note Low force contacts meet the requirements of standard force contacts

Mating and unmating

Standard insertion force contacts The maximum mating force, in pounds, shall be the number of contacts multiplied by 0 56 and the withdrawal force, in pounds, shall be a minimum of 0 08 times the number of contacts and shall not exceed the measured insertion force

Low insertion force contacts The maximum mating force, in pounds, shall be the number of contacts multiplied by 0.25 and the withdrawal force, in pounds, shall be a minimum of 0 04 times the number of contacts and shall not exceed the maximum insertion force

Contact resistance The average contact resistance of all contacts measured shall not exceed 0 010 ohm, and no individual contact pair shall have a resistance exceeding 0 020 ohm

Dielectric withstanding voltage

Sea level 1,000 volts rms

High altitude 300 volts rms

Current rating 5.0 amperes maximum per contact

Polarization Type X, regular, guide pin and socket, full round (see figure 1) Type Y, adjustable "D" shaped guide pin and/or socket (see MIL-C-55302/57, detail B) All connectors are supplied in the -1 polarized position For other than OEM use, connectors may be re-polarized by using M55302/57-01 spanner wrench and M35302/57-02 guide pin and guide socket kit When guides are removed and reinstalled in another position apply Loctite 83-21 or equivalent to the threads Ordering connectors in other than the -1 polarized position may be done by the OEM's by placing the desired polarized position in brackets three (3) spaces after the part number on the purchase order Example M55302/65-10Y -32 Polarization pins and sockets shall be adjusted to any desired configuration by spanner wrench (M55302/57-01) Ordering connectors in the -1 polarized position without Loctite 83-CV or equivalent applied may be done by placing a "-0" in brackets three (3) spaces after the part number on the purchase order Guide sets, as installed, shall withstand 7 inch ounces backoff torque applied through the guide set mounting nut

M55302/57-02 kit Consists of one type X "D" shaped guide socket and one type X "D" shaped guide pin

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(G) Insertion and removal tool MIL-I-81969/12

(G) Crimp tool M22520/2-01

Example of connector part number

M55302/65	L	10	X
Basic number of specification sheet	Type of contacts	Number of contacts (see figure 1)	Type of guide pin and/or socket (see figure 1)
	L = Low insertion force contacts - = Standard insertion force contacts	(Contacts supplied with connector are crimp removable type for no 22 maximum wire size .048 maximum insulation)	Type X = full round Type Y = "D" shaped Type F = fixed jackscrew Type S = .200 slotted turning jackscrew Type H = .200 hex turning jackscrew

CONCLUDING MATERIAL

Custodians.
Army - CR
Navy - EC
Air Force - 17
(G) NASA - NA

Review activities
Army - AR, MI
Navy - AS, TD
Air Force - 85, 99
DLA - ES

User activities
Army - AT, AV, ME
Navy - MC, OS
Air Force - 11, 19, 80

Preparing activity
Army - CR

Agent
DLA - ES

(Project 5935-3860-11)

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